EXERCISE 1: SQL FUNDAMENTALS

1.Write a SQl query to retrieve all columns from the employees table.

```
SELECT*
FROM employees;
```

id	First-name	Last _name	department	salary	Hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2.Write a SQL query to find all the unique departments in the employees table.

```
SELECT DISTINCT department

FROM Employees;
```

Departments
IT
HR
Finance
Marketing

3.Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order.

```
SELECT first_name,
        Last_name
FROM Employees
ORDER BY 'salary'  DESC;
```

First_name	Last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. Write a SQL query to retrieve the top 5 highest paid employees

```
SELECT first_name,
        Last_name,
        Salary
FROM Employees
ORDER BY salary DESC
Limit 5;
```

First_name	Last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5. write a SQL query to find employees who work in the IT department.

```
SELECT First_name,
        Last_name,
        Department
FROM Employees
WHERE Department ='IT';
```

First_name	Last_name	Department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. Write a SQL to find employees who work in the finance department AND have a salary greater than 58 000.

```
SELECT first_name,
        Last_name,
        Department,
        Salary
FROM Employees
WHERE department = 'Finance' AND salary >58000;
```

First_name	Last_name	Department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

7. Write a SQL query to find employees who work in the HR department OR the Marketing department.

```
SELECT First_name,
        Last_name,
        Department
FROM Employees
WHERE Department = 'HR' OR "Marketing';
```

First_name	Last_name	Department
Jane	Smith	HR
Jessica	Moore	HR
David	White	Marketing
Daniel	Clark	Marketing

8. Write a SQL query to find employees who do not work in the IT department.

```
SELECT First_name,
        Last_name,
        Department
FROM Employees
WHERE Department NOT 'IT";
```

First_Name	Last_name	Department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. Write a SQL query to find employees who are in the HR, IT, or Finance departments.

```
SELECT First_name,
        Last_name,
        Department
FROM Employees
WHERE Department IN ('HR','IT','FINANCE');
```

First_Name	Last_name	Department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hill	IT

10. Write a SQL query to find employees who are in the IT department, have a salary greater than 50,000, and are located in New York.

```
SELECT First_name,
        Last_name,
        Department,
        Salary,
        City
FROM Employees
WHERE department= 'IT' AND salary > 50000 AND city ='New York';
```

First_name	Last_name	Department	salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

11. Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department, earn more than 52,000, and order the results by salary in descending order.

```
SELECT First_name,
        Last_name,
        Department,
        Salary
FROM Employees
WHERE salary >52000 AND department = 'IT' OR 'Marketing'
ORDER BY salary DESC;
```

First_name	Last_name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

12. Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments.

```
SELECT DISTINCT City

FROM Employees

WHERE department NOT IN 'IT' OR' HR';
```

City
Los Angeles
San Francisco
Houston
Chicago

13. Write a SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50,000, and order the results by hire date in ascending order.

```
SELECT First_name,

        Last_name,

        Department,

        Salary,

        Hire_date

FROM Employees

WHERE department NOT IN 'Finance' AND salary> 50000
```

ORDER BY 'Hire_date' ASC;

First_name	Last_name	Department	Salary	Hire_date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

14. Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department.

```
SELECT First_name,
        Last_name,
        City,
        Department
FROM Employees
WHERE city IN 'Chicago' OR 'Los Angeles' AND department = 'IT' OR 'Marketing'
ORDER BY Hire_date ASC
Limit 3;
```

First_name	Last_name	City	Department
Emily	Davis	Chicago	IT
Daniel	Clark	Chicago	Marketing